

ALI MANSOOR

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PROFESSIONAL SUMMARY

AI/ML Engineering student (B.Tech CSE, 2027) focused on building LLM-powered and agentic AI applications using LangChain, LangGraph, and Groq. Hands-on experience building RAG pipelines with FAISS and shipping AI applications with Flask and Streamlit. Familiar with classical machine learning using scikit-learn and XGBoost, along with Docker-based development.

EDUCATION

B.Tech, Computer Science & Engineering

| 2023 - 2027

Dr. A.P.J. Abdul Kalam Technical University, Lucknow

| SGPA: 7.3/10

TECHNICAL SKILLS

Languages: Python, Java

Agentic AI LLMs: LangChain, LangGraph, RAG, Groq LLM APIs, Hugging Face Transformers

Machine Learning & Deep Learning: scikit-learn, XGBoost, TensorFlow, Keras, NLP, Feature Engineering

Libraries & Frameworks: Pandas, NumPy, Matplotlib, Seaborn, TensorFlow, OpenCV, BeautifulSoup, NLTK

Backend & APIs: FastAPI, REST APIs, Streamlit, Flask

Databases: SQL, SQLite, PostgreSQL, FAISS, Chroma

PROJECTS

Agentic Natural Language-to-SQL System | Python, LangChain, LangGraph, Groq, Streamlit

<https://github.com/aLi-mansoor04/Natural-language-to-SQL-query>

- Built a Agentic system that converts natural language queries into safe, executable SQL queries and retrieves results from relational databases using LangGraph state-graph architecture and Groq-hosted LLM.
- Engineered FAISS-based schema retrieval, self-correction loops and multi-turn conversational context to improve query accuracy on complex, multi-table schemas.
- Developed a Streamlit interface with conversational query history, generated SQL display, result visualization, JSON export, and human-in-the-loop confirmation for database write operations.

AI Multi-Agent Research System | Python, LangChain, LangGraph, Tavily, Streamlit

<https://github.com/aLi-mansoor04/ai-research-agent>

- Architected an autonomous 5-agent LLM pipeline (Search → Scrape → Write → Critique → Refine) using LangGraph for stateful, graph-based orchestration of end-to-end research report generation.
- Implemented a self-reflection loop — a Critic Agent reviews output quality and triggers iterative rewrites without human intervention — using a Groq-hosted LLM for sub-second inference latency.
- Integrated the Tavily Search API and BeautifulSoup for real-time web retrieval, and deployed a production Streamlit frontend with live agent tracking and multi-format export (TXT, Markdown), extensible to RAG and vector database retrieval.

AI-Powered RAG Chatbot | Python, LangChain, LangGraph, FAISS, Streamlit

<https://github.com/aLi-mansoor04/conversational-rag-chatbot>

- Built an LLM-powered conversational chatbot using LangChain and LangGraph with **RAG, short-term conversational memory, and FAISS-based semantic retrieval** for context-aware responses.
- Implemented a document ingestion pipeline covering **text extraction, chunking, embeddings, and vector similarity search** to retrieve relevant context before LLM generation.
- Developed an interactive Streamlit UI with conversational history and integrated Groq-hosted LLMs for responsive, context-grounded question answering.

CERTIFICATIONS

- Machine Learning Specialization — Coursera
- Deep Learning Specialization — Coursera
- **AI-Enabled Applications** — IBM SkillsBuild

ACHIEVEMENTS

- Solved 200+ Data Structures & Algorithms problems on LeetCode & HackerRank (dynamic programming, graphs, strings).
- Secured 4th position out of 166 teams in college codeathon.